

Prevalence Ratio Estimation via Logistic Regression: Extension to Continuous Covariates, Longitudinal Data, and Complex Survey Designs

Leila D. Amorim¹ and Raydonal Ospina^{1,2}

¹Department of Statistics, LInCA, Universidade Federal da Bahia,
Instituto de Matemática e Estatística, Salvador, BA, Brazil

²Department of Statistics, CASTLab, Universidade Federal de Pernambuco, Recife,
PE, Brazil

Correspondence: raydonal@de.ufpe.br; leiladen@ufba.br

Abstract. The `prLogistic` package [Amorim & Ospina, 2021] provides tools for estimating adjusted prevalence ratios (PR) from logistic regression models, overcoming the well-known bias of odds ratios (OR) for common binary outcomes. Version 2.0 introduces four methodological extensions beyond those described in the original paper: *(i)* flexible reference-value specification for continuous and polytomous covariates; *(ii)* a corrected population-averaged gradient for the marginal delta-method estimator; *(iii)* PR estimation for longitudinal binary outcomes via Generalised Estimating Equations (GEE); and *(iv)* design-consistent PR estimation for data arising from complex probability samples. We present the mathematical derivations for each extension, discuss implementation choices, and illustrate all new features with worked examples using the datasets bundled with the package.

Keywords: prevalence ratio, logistic regression, continuous covariates, generalised estimating equations, complex survey designs, R package.

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1 Introduction

The original `prLogistic` paper [Amorim & Ospina, 2021] demonstrated how logistic regression can be used to estimate adjusted prevalence ratios (PR) for independent and clustered binary outcomes, using the delta method or bootstrap for inference. The conclusion of that paper explicitly listed three directions for future work:

- (i) extension to Generalised Estimating Equations (GEE) for longitudinal data;
- (ii) handling of non-categorical (continuous) covariates; and
- (iii) integration with the `survey` package for complex probability samples.

Version 2.0 of `prLogistic` implements all three extensions, together with a methodological refinement of the marginal standardisation gradient (Section 4). This document describes the mathematical foundations of each new feature and provides complete R examples.

2 Review of the Existing Framework

We briefly recall the notation and estimators of Amorim & Ospina [2021].

2.1 The logistic model and the prevalence ratio

Let $Y_i \in \{0, 1\}$ be a binary outcome and $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ik})^\top$ the covariate vector (with intercept) for subject i . The logistic model specifies

$$P(Y_i = 1 \mid \mathbf{x}_i) = \text{expit}(\mathbf{x}_i^\top \boldsymbol{\beta}) := \frac{\exp(\mathbf{x}_i^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta})}, \quad (1)$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_k)^\top$ is estimated by maximum likelihood.

For a binary predictor $x_1 \in \{0, 1\}$, the PR comparing the exposed ($x_1 = 1$) with the unexposed ($x_1 = 0$) group is

$$\text{PR} = \frac{P(Y = 1 \mid x_1 = 1, x_2, \dots, x_k)}{P(Y = 1 \mid x_1 = 0, x_2, \dots, x_k)}. \quad (2)$$

Because PR depends on the values of the remaining covariates $(x_2, \dots, x_k)$, a standardisation procedure must be specified.

2.2 Conditional standardisation (original)

Under conditional standardisation, covariates other than x_j are fixed at reference values $\mathbf{r} = (r_2, \dots, r_k)^\top$. Let $\mathbf{x}_1 = (1, 1, r_2, \dots, r_k)^\top$ and $\mathbf{x}_0 = (1, 0, r_2, \dots, r_k)^\top$. The

conditional PR estimator is

$$\widehat{\text{PR}}_j^{\text{cond}} = \frac{\text{expit}(\mathbf{x}_1^\top \hat{\boldsymbol{\beta}})}{\text{expit}(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})}. \quad (3)$$

2.3 Marginal standardisation (original)

Under marginal standardisation, the covariate distribution of the observed sample is used. Let $\hat{\eta}_i^{(1)} = \hat{\beta}_0 + \hat{\beta}_j + \sum_{k' \neq j} \hat{\beta}_{k'} x_{ik'}$ and $\hat{\eta}_i^{(0)} = \hat{\beta}_0 + \sum_{k' \neq j} \hat{\beta}_{k'} x_{ik'}$. The marginal PR estimator is

$$\widehat{\text{PR}}_j^{\text{marg}} = \frac{\bar{p}_1}{\bar{p}_0}, \quad \bar{p}_s = n^{-1} \sum_{i=1}^n \text{expit}(\hat{\eta}_i^{(s)}), \quad s \in \{0, 1\}. \quad (4)$$

2.4 Delta-method inference (original)

Both estimators are functions of $\hat{\boldsymbol{\beta}}$. Applying the delta method to $\log(\widehat{\text{PR}})$ gives

$$\widehat{\text{Var}}[\log \widehat{\text{PR}}] \approx \mathbf{x}_*^\top \widehat{\boldsymbol{\Sigma}} \mathbf{x}_*, \quad (5)$$

where $\widehat{\boldsymbol{\Sigma}}$ is the estimated covariance matrix of $\hat{\boldsymbol{\beta}}$ and $\mathbf{x}_*$ is the gradient of $\log \widehat{\text{PR}}$ with respect to $\boldsymbol{\beta}$, defined as [Oliveira et al., 1997]

$$\mathbf{x}_* = (1 - \hat{p}_1) \mathbf{x}_1 - (1 - \hat{p}_0) \mathbf{x}_0. \quad (6)$$

The $(1 - \alpha)$ confidence interval for PR is then $\exp(\log \widehat{\text{PR}} \pm z_{\alpha/2} \widehat{\text{se}}(\log \widehat{\text{PR}}))$.

3 Extension 1: Flexible Reference Values for Continuous and Polytomous Covariates

3.1 The problem with binary-only reference values

The original implementation hard-coded the reference value of every predictor as zero. This is appropriate for binary (0/1) indicators but produces uninformative or impossible reference profiles for continuous variables (e.g. age, body weight) or for dummy-coded polytomous factors.

3.2 The generalised conditional PR estimator

Let $\mathbf{r} = (r_1, \dots, r_k)^\top$ be a vector of reference values, one per predictor. For predictor j , define

$$\mathbf{x}_1^{(j)} = (1, r_1, \dots, r_{j-1}, 1, r_{j+1}, \dots, r_k)^\top, \quad (7)$$

$$\mathbf{x}_0^{(j)} = (1, r_1, \dots, r_{j-1}, 0, r_{j+1}, \dots, r_k)^\top. \quad (8)$$

All other predictors are fixed at their reference values $r_{k'}$; only predictor j is contrasted at 1 versus 0.

Remark 1. *For a continuous predictor x_j , the “exposed” value is 1 and the “unexposed” value is 0. This corresponds to a one-unit increase on the original scale (or a clinically meaningful increment if the predictor is centred or scaled). The analyst should select $\mathbf{r}$ at a substantively meaningful point — for instance, the sample median or mean — and interpret $\widehat{\text{PR}}_j$ as the PR associated with a one-unit increase starting from r_j .*

3.3 Reference value selection rules

prLogistic v2.0 applies the following automatic defaults:

Predictor type	Detection	Default r_j
Binary indicator	all values in $\{0, 1\}$	0
Continuous / count	otherwise	sample median

The user may override any r_j through the `ref_values` argument, e.g. `ref_values = list(age = 40, bmi = 25)`. User-supplied values always take precedence.

3.4 Revised delta-method gradient

With the generalised reference rows (7)–(8), the gradient (6) becomes

$$\mathbf{x}_*^{(j)} = (1 - \hat{p}_1^{(j)}) \mathbf{x}_1^{(j)} - (1 - \hat{p}_0^{(j)}) \mathbf{x}_0^{(j)}, \quad (9)$$

where $\hat{p}_s^{(j)} = \text{expit}(\mathbf{x}_s^{(j)\top} \hat{\boldsymbol{\beta}})$, $s \in \{0, 1\}$. The variance estimator and confidence interval follow from (5) applied with $\mathbf{x}_* = \mathbf{x}_*^{(j)}$.

3.5 R usage

```
1 data(LBW, package = "prLogistic")
2 LBW$low_bin <- as.integer(LBW$low == "Low")
```

```

3
4 fit <- glm(low_bin ~ as.integer(smoke == "Yes") + age,
5           family = binomial, data = LBW)
6
7 # Default: age at its sample median
8 prLogisticDelta(fit, standardisation = "conditional")
9
10 # Custom reference: age = 25 years
11 prLogisticDelta(fit,
12                 standardisation = "conditional",
13                 ref_values      = list(age = 25))
14
15 # Reference continuous covariates at mean instead of median
16 prLogisticDelta(fit,
17                 standardisation = "conditional",
18                 ref_continuous  = "mean")

```

Listing 1: Continuous covariates with custom reference values

4 Extension 2: Corrected Marginal Delta-Method Gradient

4.1 The original approximation and its limitation

The original implementation approximated the gradient of $\log(\widehat{\text{PR}}^{\text{marg}})$ using a single pair of contrast vectors evaluated at the marginal means of the covariates. This is exact only when the model is linear in the covariates and underestimates the gradient in the presence of non-linearity (as induced by the logistic link).

4.2 Corrected population-averaged gradient

The correct gradient of $\log(\widehat{\text{PR}}_j^{\text{marg}})$ with respect to β is obtained by the chain rule:

$$\frac{\partial \log \widehat{\text{PR}}_j^{\text{marg}}}{\partial \beta} = \frac{1}{\bar{p}_1} \frac{\partial \bar{p}_1}{\partial \beta} - \frac{1}{\bar{p}_0} \frac{\partial \bar{p}_0}{\partial \beta}. \quad (10)$$

Since $\bar{p}_s = n^{-1} \sum_i \text{expit}(\hat{\eta}_i^{(s)})$ and $\partial \text{expit}(\eta) / \partial \beta = \text{expit}(\eta) [1 - \text{expit}(\eta)] \mathbf{x}_i^{(s)}$, we have

$$\frac{\partial \bar{p}_s}{\partial \beta} = n^{-1} \sum_{i=1}^n \hat{p}_{is} (1 - \hat{p}_{is}) \mathbf{x}_i^{(s)}, \quad (11)$$

where $\hat{p}_{is} = \text{expit}(\hat{\eta}_i^{(s)})$ and $\mathbf{x}_i^{(s)}$ is the counterfactual design row of subject i when predictor j is set to $s \in \{0, 1\}$.

Substituting (11) into (10) and using $1/\bar{p}_s \approx (n^{-1} \sum_i \hat{p}_{is})^{-1}$, the delta-method gradient simplifies to

$$\mathbf{x}_*^{\text{marg}} = \overline{(1 - \hat{p}_{i1})\mathbf{x}_i^{(1)} - (1 - \hat{p}_{i0})\mathbf{x}_i^{(0)}} \quad (12)$$

where $\overline{(\cdot)}$ denotes the sample mean over all n subjects. The variance estimator is then

$$\widehat{\text{Var}}\left[\log \widehat{\text{PR}}_j^{\text{marg}}\right] \approx \mathbf{x}_*^{\text{marg}\top} \widehat{\Sigma} \mathbf{x}_*^{\text{marg}}. \quad (13)$$

Remark 2. Expression (12) is the empirical influence function of $\log \widehat{\text{PR}}^{\text{marg}}$ averaged over the sample [Hampel et al., 1986]. It collapses to the original gradient (6) in the special case where every subject has the same covariate profile (i.e., all $\mathbf{x}_i^{(s)}$ are identical), confirming that the correction has no effect for balanced binary designs but matters for heterogeneous continuous predictors.

5 Extension 3: Generalised Estimating Equations (GEE)

5.1 Motivation

Longitudinal and repeated-measures studies produce clustered binary outcomes where observations within the same unit (patient, household, school) are correlated. Random-effects logistic models (Section 1.3 of Amorim & Ospina [2021]) yield *subject-specific* (conditional) parameters. Generalised Estimating Equations [Zeger & Liang, 1986] instead target *population-averaged* (marginal) parameters, which are more natural for prevalence ratio interpretation.

5.2 The GEE model

Let $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{in_i})^\top$ be the vector of binary responses for cluster i , $i = 1, \dots, m$. GEE specifies the marginal mean model

$$E(Y_{it} \mid \mathbf{x}_{it}) = \mu_{it} = \text{expit}(\mathbf{x}_{it}^\top \boldsymbol{\beta}), \quad (14)$$

where $\mathbf{x}_{it}$ is the covariate vector at occasion t , and estimates $\boldsymbol{\beta}$ by solving the GEE score equations

$$\sum_{i=1}^m \mathbf{D}_i^\top \mathbf{V}_i^{-1} (\mathbf{Y}_i - \boldsymbol{\mu}_i) = \mathbf{0}, \quad (15)$$

where $\mathbf{D}_i = \partial \boldsymbol{\mu}_i / \partial \boldsymbol{\beta}^\top$ and $\mathbf{V}_i = \mathbf{A}_i^{1/2} \mathbf{R}(\alpha) \mathbf{A}_i^{1/2}$ is a working covariance matrix constructed from the working correlation matrix $\mathbf{R}(\alpha)$ and the diagonal matrix of marginal variances $\mathbf{A}_i = \text{diag}(\mu_{it}[1 - \mu_{it}])$ [Liang & Zeger, 1986].

5.3 Robust variance and PR estimation

A key property of GEE is that the *sandwich* (robust) variance estimator

$$\widehat{\text{Var}}_{\text{sand}}(\hat{\boldsymbol{\beta}}) = \mathbf{B}^{-1} \mathbf{M} \mathbf{B}^{-1}, \quad (16)$$

where $\mathbf{B} = \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i$ and $\mathbf{M} = \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} (\mathbf{Y}_i - \boldsymbol{\mu}_i)(\mathbf{Y}_i - \boldsymbol{\mu}_i)^\top \mathbf{V}_i^{-1} \mathbf{D}_i$, is consistent for the true variance of $\hat{\boldsymbol{\beta}}$ regardless of whether $\mathbf{R}(\alpha)$ is correctly specified [Liang & Zeger, 1986].

`prLogistic` v2.0 extracts $\hat{\boldsymbol{\Sigma}} = \widehat{\text{Var}}_{\text{sand}}(\hat{\boldsymbol{\beta}})$ directly from the `geese$vbeta` slot of a `geeglm()` object (from the `geepack` package), bypassing the fragile `vcov.geeglm` call that fails in older `geepack` versions. PR estimates are then computed from (3) or (4) using $\hat{\boldsymbol{\beta}}$ and the robust $\hat{\boldsymbol{\Sigma}}$ in (5).

Remark 3. *Because GEE targets marginal parameters, marginal standardisation (Section 2.3) is the natural and recommended choice. Conditional standardisation is available for completeness but yields a conditional-on-working-correlation interpretation.*

5.4 Working correlation structures

corstr	Structure $\mathbf{R}(\alpha)$	Suitable for
"independence"	$\mathbf{I}$	No repeated measures
"exchangeable"	$(1 - \alpha)\mathbf{I} + \alpha \mathbf{1}\mathbf{1}^\top$	Equal time-gaps
"ar1"	$R_{ts} = \alpha^{ t-s }$	Ordered time
"unstructured"	R_{ts} unconstrained	Few time points

5.5 R usage

```

1 library(geepack)
2 data(Toenail, package = "prLogistic")
3 Toenail$resp_bin <- as.integer(Toenail$Response == "Moderate/
  severe")
4 Toenail$trt_bin <- as.integer(Toenail$Treatment == "
  Terbinafine")
5
6 fit_gee <- geeglm(

```



```

7   resp_bin ~ trt_bin + Month,
8   family = binomial,
9   id      = ID,
10  corstr = "exchangeable",
11  data    = Toenail
12 )
13
14 # Population-averaged PR (marginal, recommended for GEE)
15 prLogisticGEE(fit_gee)
16
17 # With bootstrap confidence intervals
18 prLogisticGEE(fit_gee,
19               method = "bootstrap",
20               data    = Toenail,
21               R        = 999)

```

Listing 2: GEE for longitudinal toenail infection data

6 Extension 4: Complex Survey Designs

6.1 The design problem

Nationally representative health and epidemiological surveys use stratified multistage cluster sampling (SMCS). Analysing such data as if it were a simple random sample produces biased point estimates and inconsistent standard errors. The **survey** package [Lumley, 2004] implements design-consistent estimation for a broad class of statistics, including logistic regression via `svyglm()`.

6.2 Design-consistent logistic regression

Let the survey design be characterised by stratum indicators h , cluster indicators c , and sampling weights w_i . The pseudo-maximum-likelihood estimator for β solves the weighted score equation

$$\sum_h \sum_{c \in h} \sum_{i \in c} w_i \mathbf{x}_i (Y_i - \mu_i) = \mathbf{0}, \quad (17)$$

which incorporates the survey weights into the estimation.

The linearisation (Taylor series) variance estimator for $\hat{\beta}$ accounts for stratification,

clustering, and finite-population corrections:

$$\widehat{\text{Var}}_{\text{lin}}(\hat{\beta}) = \mathbf{B}_w^{-1} \widehat{\text{Var}} \left(\sum_{h,c,i} w_i \mathbf{x}_i \tilde{e}_i \right) \mathbf{B}_w^{-\top}, \quad (18)$$

where $\tilde{e}_i = Y_i - \hat{\mu}_i$ are the Pearson residuals weighted by the inverse of the expected information matrix $\mathbf{B}_w = \sum w_i \hat{\mu}_i (1 - \hat{\mu}_i) \mathbf{x}_i \mathbf{x}_i^\top$.

`prLogistic v2.0` extracts $\hat{\Sigma} = \widehat{\text{Var}}_{\text{lin}}(\hat{\beta})$ via the standard `vcov()` method of a `svyglm()` object and uses it in place of the ordinary MLE covariance matrix in (5). No modification to the PR estimators (3)–(4) is required; the design information enters exclusively through $\hat{\Sigma}$.

Remark 4. *Bootstrap confidence intervals for survey data require design-aware resampling (e.g., the `survey` bootstrap or balanced repeated replication). `prLogistic v2.0` provides the delta-method confidence interval for `svyglm()` objects. Bootstrap resampling can be performed using the bootstrap-replicate survey design functionality of the `survey` package, which is beyond the scope of this document.*

6.3 R usage

```

1 library(survey)
2 data(api, package = "survey")
3
4 # Create binary outcome
5 apiclus2$target_met <- as.integer(apiclus2$sch.wide == "Yes")
6
7 # Define the two-stage cluster design
8 dclus2 <- svydesign(
9   id    = ~dnum + snum,
10  fpc    = ~fpc1 + fpc2,
11  data   = apiclus2
12 )
13
14 # Fit design-consistent logistic regression
15 fit_svy <- svyglm(
16   target_met ~ meals + stype,
17   design = dclus2,
18   family = quasibinomial
19 )
20
21 # Design-consistent PR (conditional)

```

```

22 prLogisticSurvey(fit_svy, standardisation = "conditional")
23
24 # Design-consistent PR (marginal)
25 prLogisticSurvey(fit_svy, standardisation = "marginal")

```

Listing 3: Design-consistent PR for a two-stage cluster sample

7 New S3 Class and Methods

All estimation functions now return an object of class "prLogistic" with components listed in Table 1.

Table 1: Components of a prLogistic object.

Component	Type	Description
table	matrix	PR estimates and CI bounds ($p \times 3$ or $p \times 5$)
conf	numeric	Confidence level
method	character	"delta" or "bootstrap"
standardisation	character	"conditional" or "marginal"
model_type	character	"glm", "glmerMod", "geeglm", or "svyglm"
call	call	The matched function call

The following S3 methods are available:

```

1 res <- prLogisticDelta(fit, standardisation = "conditional")
2
3 print(res)           # formatted summary table
4 summary(res)         # summary + call
5 coef(res)            # named vector of PR point estimates
6 confint(res)         # CI matrix (type = "normal" or "percentile"
   " for bootstrap)
7 plot(res)            # base-R forest plot

```

Listing 4: S3 methods for prLogistic objects

8 Comparison of All Supported Model Types

Table 2 summarises the four model types, their assumptions, and the appropriate standardisation choice.

Table 2: Model types supported by `prLogistic` v2.0.

Function	Class	Covariance $\hat{\Sigma}$	Assumption	Recommended std.
<code>prLogisticDelta()</code> (<code>glm</code>)	<code>glm</code>	MLE information matrix	Independent observations	Conditional or marginal
<code>prLogisticDelta()</code> (<code>glmerMod</code>)	<code>glmerMod</code>	Fixed-effect block of MLE information	i.i.d. random effects	Marginal
<code>prLogisticGEE()</code>	<code>geeglm</code>	Sandwich $\widehat{\text{Var}}_{\text{sand}}$	Within-cluster correlation	Marginal
<code>prLogisticSurvey()</code>	<code>svyglm</code>	Linearisation $\widehat{\text{Var}}_{\text{lin}}$	Complex probability sample	Conditional or marginal

9 Illustrative Example: Low Birth Weight (LBW)

We use the LBW dataset (included in `prLogistic`), a longitudinal study of $n = 244$ mothers followed through two pregnancies in Salvador, Brazil. The binary outcome is low birth weight ($< 2,500$ g). Predictors include maternal smoking (`smoke`), race (`race`), and age at delivery (`age`, a centred continuous variable). The dataset has 488 observations (two births per mother), so clustering must be accounted for.

Table 3 reports PR estimates for maternal smoking across all four estimation approaches.

Table 3: Estimated PR and 95% CI for maternal smoking on low birth weight (LBW dataset), four estimation approaches. All models include race and age as covariates.

Model	Standardisation	PR	Lower 95%	Upper 95%
GLM (independent)	Conditional	1.74	1.28	2.37
GLM (independent)	Marginal	1.71	1.27	2.31
GLMM (random intercept)	Marginal	1.69	1.24	2.30
GEE (exchangeable)	Marginal	1.72	1.27	2.33
OR (logistic, GLM)	—	2.30	1.51	3.51

The OR (2.30) substantially overstates the PR (≈ 1.70 – 1.74), reflecting the 18% prevalence of low birth weight in this sample. All four PR approaches agree closely, confirming robustness across modelling assumptions.

```

1 library(prLogistic)
2 library(geepack)
3 library(lme4)
4
5 data(LBW, package = "prLogistic")
6 LBW$low_bin <- as.integer(LBW$low == "Low")
7 LBW$smoke_bin <- as.integer(LBW$smoke == "Yes")
8 LBW$race_bin <- as.integer(LBW$race == "Non-white")
9

```

```

10 # GLM (ignores clustering)
11 fit_glm <- glm(low_bin ~ smoke_bin + race_bin + age,
12               family = binomial, data = LBW)
13 prLogisticDelta(fit_glm, "conditional")
14 prLogisticDelta(fit_glm, "marginal")
15
16 # GLMM (random intercept per mother)
17 fit_ml <- glmer(low_bin ~ smoke_bin + race_bin + age + (1 | ID
18               ),
19               family = binomial, data = LBW)
20 prLogisticDelta(fit_ml, "marginal")
21
22 # GEE (exchangeable correlation between births)
23 fit_gee <- geeglm(low_bin ~ smoke_bin + race_bin + age,
24                  family = binomial, id = ID,
25                  corstr = "exchangeable", data = LBW)
26 prLogisticGEE(fit_gee)

```

Listing 5: Reproducing Table 2 in R

10 Discussion and Conclusion

`prLogistic` v2.0 completes the research agenda set out in Amorim & Ospina [2021] by implementing the three extensions identified as future work: GEE for longitudinal data, continuous covariates, and integration with the `survey` package. Together with the corrected marginal delta-method gradient and the new S3 class, these additions make `prLogistic` a comprehensive tool for adjusted PR estimation across the data structures most commonly encountered in public health and epidemiological research.

Key practical recommendations:

- (i) Always use OR only as an approximation to PR when prevalence $< 10\%$; for more common outcomes, use `prLogistic`.
- (ii) Prefer *marginal* standardisation for public-health policy questions (population-averaged effect); prefer *conditional* for clinical or mechanistic questions.
- (iii) For longitudinal data, use `prLogisticGEE()` with `corstr = "exchangeable"` or `"ar1"` as the working correlation structure.
- (iv) For survey data, fit `svyglm()` first with the correct design object, then pass the result to `prLogisticSurvey()`.

- (v) Use bootstrap confidence intervals as a sensitivity check in small samples ($n < 100$) or extreme prevalences.

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